

• Borel Cantelli Lemma

Consider a sequence of random variables $X_1, X_2, \dots$

If for all ϵ ,

$$\sum_{n=1}^{\infty} P(|X_n - X| > \epsilon) < \infty$$

Then $X_n \rightarrow X$ a.s.

sufficient condition for a.s. convergence, not necessary.

$$\sum_{n=1}^{\infty} P(|X_n - X| > \epsilon) = E \left[\sum_{n=1}^{\infty} 1_{\{|X_n - X| > \epsilon\}} \right]$$

$$E[1_A] = P(A)$$

NOTE: Just because each $P(|X_n - X| > \epsilon)$ is finite, does not mean their summation is finite.

* Examples given in ppt

• Convergence in probability

X_n converges to X in probability if:

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0 \quad \text{for all } \epsilon > 0$$

a.s. convergence implies convergence in probability.

• Convergence in r th mean

X_n converges to X in r th mean if

$$\lim_{n \rightarrow \infty} E[|X_n - X|^r] = 0$$

convergence in r th mean implies convergence in probability.

• Convergence in distribution / weak convergence

X_n converges to X in distribution if:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \text{for all continuity points of } F_X(\cdot).$$

convergence in probability implies convergence in distribution.

a.s. convergence

convergence in
 r th mean

convergence in probability

convergence in distribution